

Mathematical Foundations for FFT-Based AI Image Detection

From the Discrete Fourier Transform to
Polar Feature Extraction and Statistical Learning
Charitha Ranasinghe
20.04.2026

Contents

1	The Discrete Fourier Transform	3
1.1	One-Dimensional DFT	3
1.2	Two-Dimensional DFT	3
1.3	Separability	4
2	Frequency Shifting and the DC Component	4
2.1	Zero-Frequency (DC) Component	4
2.2	FFT Shift — Centring the Spectrum	4
3	Magnitude Spectrum and Log Scaling	4
4	Power Spectrum and Radial Analysis	5
4.1	Power Spectral Density	5
4.2	DC Energy Fraction	5
4.3	Radial Power Profile	5
5	Polar Frequency Feature Extraction	6
5.1	Polar Binning	6
5.2	Scalar Summary Features	6
6	Residual Peak Detection (Grid Artifact Test)	7
7	Feature Standardization	8
8	Classification Models	8
8.1	Logistic Regression	8
8.2	Random Forest	8
8.3	Gradient Boosting	9
9	Model Evaluation	9

9.1	ROC Curve and AUC	9
9.2	5-Fold Cross-Validation	9
10	Summary of Feature Vector	10

1 The Discrete Fourier Transform

1.1 One-Dimensional DFT

Let $\mathbf{x} = (x_0, x_1, \dots, x_{N-1}) \in \mathbb{C}^N$ be a finite sequence of length N .

1-D DFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1. \quad (1)$$

The corresponding *inverse* DFT is:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{+j2\pi kn/N}. \quad (2)$$

Here $j = \sqrt{-1}$, and $W_N = e^{-j2\pi/N}$ is the primitive N -th root of unity. The basis functions $\{e^{j2\pi kn/N}\}$ are orthogonal:

$$\sum_{n=0}^{N-1} e^{j2\pi mn/N} e^{-j2\pi kn/N} = N \delta_{mk}, \quad (3)$$

where δ_{mk} is the Kronecker delta.

1.2 Two-Dimensional DFT

A grayscale image is a matrix $\mathbf{I} \in \mathbb{R}^{H \times W}$, where $I[m, n]$ is the intensity at pixel (m, n) . The **2-D DFT** decomposes the image into a superposition of 2-D complex exponentials (spatial frequency components):

2-D DFT

$$F[u, v] = \sum_{m=0}^{H-1} \sum_{n=0}^{W-1} I[m, n] e^{-j2\pi(\frac{mu}{H} + \frac{nv}{W})}, \quad (4)$$

for spatial frequencies $(u, v) \in \{0, \dots, H-1\} \times \{0, \dots, W-1\}$. The pair (u, v) encodes cycles per image in the vertical and horizontal directions respectively.

The inverse transform recovers the image:

$$I[m, n] = \frac{1}{HW} \sum_{u=0}^{H-1} \sum_{v=0}^{W-1} F[u, v] e^{+j2\pi(\frac{mu}{H} + \frac{nv}{W})}. \quad (5)$$

1.3 Separability

Because the exponent separates multiplicatively, the 2-D DFT is *separable*: it equals the 1-D DFT applied along rows, followed by the 1-D DFT applied along columns:

$$F[u, v] = \underbrace{\sum_{m=0}^{H-1} e^{-j2\pi mu/H}}_{\text{column DFT of row DFTs}} \overbrace{\left(\sum_{n=0}^{W-1} I[m, n] e^{-j2\pi nv/W} \right)}^{\text{row DFT}}. \quad (6)$$

This separability is precisely what the Fast Fourier Transform (FFT) algorithm exploits to reduce complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \log N)$.

2 Frequency Shifting and the DC Component

2.1 Zero-Frequency (DC) Component

The coefficient at $(u, v) = (0, 0)$ is called the *DC component*:

$$F[0, 0] = \sum_{m=0}^{H-1} \sum_{n=0}^{W-1} I[m, n] = HW \bar{I}, \quad (7)$$

where $\bar{I}$ denotes the mean pixel intensity. It carries the total *energy bias* of the image.

2.2 FFT Shift — Centring the Spectrum

The raw DFT output places the DC component at the corner $(0, 0)$. Applying the frequency-shift theorem via multiplication by $(-1)^{m+n}$:

$$I[m, n] \cdot (-1)^{m+n} \xleftrightarrow{\mathcal{F}_{2D}} F\left[u - \frac{H}{2}, v - \frac{W}{2}\right], \quad (8)$$

translates the DC component to the geometric centre $(\frac{H}{2}, \frac{W}{2})$, making the spectrum visually interpretable. We denote the shifted spectrum

$$\hat{F}[u, v] = F\left[u - \frac{H}{2}, v - \frac{W}{2}\right]. \quad (9)$$

3 Magnitude Spectrum and Log Scaling

Since $\hat{F}[u, v] \in \mathbb{C}$, we define the **magnitude spectrum**:

$$\mathcal{M}[u, v] = \left| \hat{F}[u, v] \right| = \sqrt{\text{Re}(\hat{F}[u, v])^2 + \text{Im}(\hat{F}[u, v])^2}. \quad (10)$$

The dynamic range of $\mathcal{M}$ spans many orders of magnitude (the DC coefficient can be $10^6 \times$ larger than high-frequency coefficients). Visualization and feature extraction therefore use **log-compression**:

Log Magnitude Spectrum

$$\mathcal{L}[u, v] = 20 \log(1 + |\hat{F}[u, v]|) \approx 20 \log |\hat{F}[u, v]|, \quad (11)$$

where $\log$ denotes the natural logarithm and the $+1$ is a numerical stabilizer that prevents $\log(0)$. The factor 20 converts to a decibel-like scale.

Remark 3.1. The phase spectrum $\varphi[u, v] = \angle \hat{F}[u, v] = \arctan(\text{Im} / \text{Re})$ encodes the spatial alignment of features but carries less discriminative information for the natural-vs-AI classification task.

4 Power Spectrum and Radial Analysis

4.1 Power Spectral Density

The **power spectrum** (periodogram) is

$$\mathcal{P}[u, v] = |\hat{F}[u, v]|^2. \quad (12)$$

By Parseval’s theorem the total power in the spatial domain equals the total power in the frequency domain:

$$\sum_{m,n} |I[m, n]|^2 = \frac{1}{HW} \sum_{u,v} |F[u, v]|^2. \quad (13)$$

4.2 DC Energy Fraction

DC Energy Fraction

$$\eta_{\text{DC}} = \frac{\mathcal{P}[\frac{H}{2}, \frac{W}{2}]}{\sum_{u=0}^{H-1} \sum_{v=0}^{W-1} \mathcal{P}[u, v]}. \quad (14)$$

Natural images obey a *pink-noise* (“ $1/f$ ”) power law, concentrating most energy near DC. AI-generated images distribute energy more uniformly across frequencies, so η_{DC} is a primary discriminative feature.

4.3 Radial Power Profile

Define polar coordinates centred at the DC position:

$$r[m, n] = \sqrt{(n - \frac{W}{2})^2 + (m - \frac{H}{2})^2}, \quad \theta[m, n] = \arctan\left(\frac{m - \frac{H}{2}}{n - \frac{W}{2}}\right). \quad (15)$$

The **radial (azimuthal) average** at radius ρ is:

$$\bar{\mathcal{P}}(\rho) = \frac{1}{|\mathcal{A}_\rho|} \sum_{(m,n) : [r[m,n]] = \rho} \mathcal{P}[m, n], \quad (16)$$

where $\mathcal{A}_\rho$ is the set of pixels at integer radius ρ .

For natural photographs the radial profile follows $\bar{\mathcal{P}}(\rho) \propto \rho^{-\alpha}$ with $\alpha \approx 2$ (the classical $1/f^2$ spectral law of natural images). AI-generated images deviate from this monotone decay, exhibiting bumps or plateaus at specific radii.

5 Polar Frequency Feature Extraction

5.1 Polar Binning

Let $R_{\max} = \min(\frac{H}{2}, \frac{W}{2})$ be the largest representable radius, N_r the number of radial rings, and N_s the number of angular sectors. Define:

$$\Delta r = \frac{R_{\max}}{N_r}, \quad (17)$$

$$\Delta \theta = \frac{2\pi}{N_s}. \quad (18)$$

Each **polar bin** is the set of spectrum pixels falling in ring i and sector j :

$$\mathcal{B}_{ij} = \{(m, n) \mid i \Delta r \leq r[m, n] < (i+1)\Delta r, j \Delta \theta \leq \theta[m, n] < (j+1)\Delta \theta\}. \quad (19)$$

Polar Bin Feature

$$\phi_{ij} = \frac{1}{|\mathcal{B}_{ij}|} \sum_{(m,n) \in \mathcal{B}_{ij}} \mathcal{L}[m, n], \quad i \in \{0, \dots, N_r - 1\}, j \in \{0, \dots, N_s - 1\}. \quad (20)$$

With $N_r = 32$ rings and $N_s = 16$ sectors this yields a **512-dimensional feature vector** $\phi \in \mathbb{R}^{N_r N_s}$.

5.2 Scalar Summary Features

Three additional statistics are appended:

DC Energy Fraction

Defined in (14).

Spectral Flatness (Wiener Entropy)

The **spectral flatness measure** quantifies how noise-like (flat) the power spectrum is:

Spectral Flatness

$$\gamma = \frac{\exp\left(\frac{1}{HW} \sum_{u,v} \ln(\mathcal{P}[u, v] + \varepsilon)\right)}{\frac{1}{HW} \sum_{u,v} \mathcal{P}[u, v]}, \quad (21)$$

where $\varepsilon > 0$ is a small constant for numerical stability. Pure white noise gives $\gamma = 1$; a tonal (single-frequency) signal gives $\gamma \rightarrow 0$. The numerator is the geometric mean and the denominator is the arithmetic mean of the power spectrum.

High-to-Low Frequency Energy Ratio

Partition frequency space at radius threshold $\rho_0 = 0.3 R_{\max}$:

$$E_{\text{LF}} = \frac{1}{|\Omega_{\text{in}}|} \sum_{(m,n) \in \Omega_{\text{in}}} \mathcal{P}[m, n], \quad \Omega_{\text{in}} = \{(m, n) : r[m, n] < \rho_0\}, \quad (22)$$

$$E_{\text{HF}} = \frac{1}{|\Omega_{\text{out}}|} \sum_{(m,n) \in \Omega_{\text{out}}} \mathcal{P}[m, n], \quad \Omega_{\text{out}} = \{(m, n) : r[m, n] \geq \rho_0\}. \quad (23)$$

$$\lambda = \frac{E_{\text{HF}}}{E_{\text{LF}}}. \quad (24)$$

The complete feature vector is therefore:

$$\mathbf{x} = (\phi_{00}, \dots, \phi_{(N_r-1)(N_s-1)}, \eta_{\text{DC}}, \gamma, \lambda) \in \mathbb{R}^{N_r N_s + 3}. \quad (25)$$

With $N_r = 32, N_s = 16$ this gives $\mathbf{x} \in \mathbb{R}^{515}$.

6 Residual Peak Detection (Grid Artifact Test)

To detect the periodic grid artifacts introduced by AI generators (caused by convolutional strides and attention tile sizes), the DC lobe is first removed. Define the DC-exclusion mask with radius ρ_{dc} :

$$\mathcal{L}_{\text{res}}[m, n] = \begin{cases} 0 & \text{if } r[m, n] \leq \rho_{\text{dc}}, \\ \mathcal{L}[m, n] & \text{otherwise.} \end{cases} \quad (26)$$

Let Q_τ denote the τ -th quantile of $\{\mathcal{L}_{\text{res}}[m, n] : \mathcal{L}_{\text{res}}[m, n] > 0\}$. The **peak count** is:

$$\mathcal{K} = |\{(m, n) \mid \mathcal{L}_{\text{res}}[m, n] > Q_{0.80}\}|. \quad (27)$$

Large $\mathcal{K}$ (empirically $> 0.01 HW$) indicates scattered, periodic high-energy peaks characteristic of AI generation.

7 Feature Standardization

Before training, each feature dimension k is **z-score standardized** over the training set $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$:

$$\tilde{x}_k^{(i)} = \frac{x_k^{(i)} - \mu_k}{\sigma_k}, \quad \mu_k = \frac{1}{n} \sum_{i=1}^n x_k^{(i)}, \quad \sigma_k = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_k^{(i)} - \mu_k)^2}. \quad (28)$$

8 Classification Models

8.1 Logistic Regression

Given standardised features $\tilde{\mathbf{x}} \in \mathbb{R}^d$, the probability of class $y = 1$ (AI-generated) is:

$$P(y = 1 \mid \tilde{\mathbf{x}}) = \sigma(\mathbf{w}^\top \tilde{\mathbf{x}} + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}. \quad (29)$$

Parameters $(\mathbf{w}, b)$ are found by minimizing the regularized *cross-entropy loss*:

$$\mathcal{L}(\mathbf{w}, b) = -\frac{1}{n} \sum_{i=1}^n \left[y^{(i)} \log \hat{p}^{(i)} + (1 - y^{(i)}) \log(1 - \hat{p}^{(i)}) \right] + \frac{\lambda}{2} \|\mathbf{w}\|^2, \quad (30)$$

where $\hat{p}^{(i)} = P(y = 1 \mid \tilde{\mathbf{x}}^{(i)})$ and λ controls L_2 regularization.

8.2 Random Forest

A random forest is an ensemble of T decision trees $\{h_t(\tilde{\mathbf{x}})\}_{t=1}^T$, each trained on a **bootstrap sample** $\mathcal{D}_t \sim \mathcal{D}$ and using a random feature subset of size $\lfloor \sqrt{d} \rfloor$ at each split. The class-probability estimate is the average vote:

$$\hat{P}(y = 1 \mid \tilde{\mathbf{x}}) = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[h_t(\tilde{\mathbf{x}}) = 1]. \quad (31)$$

Each tree is grown by recursively partitioning at the feature k^* and threshold τ^* that minimise the **Gini impurity**:

$$\text{Gini}(\mathcal{S}) = 1 - \sum_{c \in \{0,1\}} p_c^2, \quad (k^*, \tau^*) = \underset{k, \tau}{\arg\min} \left[\frac{|\mathcal{S}_L|}{|\mathcal{S}|} \text{Gini}(\mathcal{S}_L) + \frac{|\mathcal{S}_R|}{|\mathcal{S}|} \text{Gini}(\mathcal{S}_R) \right], \quad (32)$$

where $\mathcal{S}_L, \mathcal{S}_R$ are the left/right splits.

Feature importance of feature k is the mean decrease in Gini impurity across all trees:

$$\text{MDA}(k) = \frac{1}{T} \sum_{t=1}^T \sum_{\text{node } s \text{ splits on } k} \frac{|\mathcal{S}_s|}{|\mathcal{D}|} \left[\text{Gini}(\mathcal{S}_s) - \frac{|\mathcal{S}_{s,L}|}{|\mathcal{S}_s|} \text{Gini}(\mathcal{S}_{s,L}) - \frac{|\mathcal{S}_{s,R}|}{|\mathcal{S}_s|} \text{Gini}(\mathcal{S}_{s,R}) \right]. \quad (33)$$

8.3 Gradient Boosting

Gradient boosting constructs an additive model $F_T(\tilde{\mathbf{x}}) = \sum_{t=1}^T \nu h_t(\tilde{\mathbf{x}})$, where ν is the learning rate and each tree h_t fits the *negative gradient* (pseudo-residuals) of the loss at the current model F_{t-1} :

$$r_i^{(t)} = - \left[\frac{\partial}{\partial F(\tilde{\mathbf{x}}^{(i)})} \mathcal{L}(y^{(i)}, F(\tilde{\mathbf{x}}^{(i)})) \right]_{F=F_{t-1}}. \quad (34)$$

For the binary log-loss $\mathcal{L}(y, p) = -y \log p - (1 - y) \log(1 - p)$ the pseudo-residuals reduce to $r_i^{(t)} = y^{(i)} - \sigma(F_{t-1}(\tilde{\mathbf{x}}^{(i)}))$.

9 Model Evaluation

9.1 ROC Curve and AUC

For a classifier with score $s(\mathbf{x})$ and decision threshold δ , define the true and false positive rates:

$$\text{TPR}(\delta) = \frac{TP(\delta)}{TP(\delta) + FN(\delta)}, \quad \text{FPR}(\delta) = \frac{FP(\delta)}{FP(\delta) + TN(\delta)}. \quad (35)$$

The **Receiver Operating Characteristic (ROC) curve** traces $(\text{FPR}(\delta), \text{TPR}(\delta))$ for δ sweeping over all score values. The **Area Under the ROC Curve (AUC)** is:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t)) dt = P(s(\mathbf{x}^+) > s(\mathbf{x}^-)), \quad (36)$$

where $\mathbf{x}^+$ is a random AI sample and $\mathbf{x}^-$ a natural one.

9.2 5-Fold Cross-Validation

The training set is randomly partitioned into $K = 5$ folds $\mathcal{D}_1, \dots, \mathcal{D}_5$. For fold k the model is trained on $\mathcal{D} \setminus \mathcal{D}_k$ and evaluated on $\mathcal{D}_k$:

$$\widehat{\text{AUC}}_{\text{CV}} = \frac{1}{K} \sum_{k=1}^K \text{AUC}_k, \quad \widehat{\sigma}_{\text{CV}} = \sqrt{\frac{1}{K-1} \sum_{k=1}^K (\text{AUC}_k - \widehat{\text{AUC}}_{\text{CV}})^2}. \quad (37)$$

10 Summary of Feature Vector

Component	Symbol	Dim.	Equation
Polar bin log-power	ϕ_{ij}	512	(20)
DC energy fraction	η_{DC}	1	(14)
Spectral flatness	γ	1	(21)
HF / LF energy ratio	λ	1	(24)
Total		515	(25)

Table 1: Components of the FFT feature vector $\mathbf{x} \in \mathbb{R}^{515}$.